Digital Payments Transaction Analytics

Segmentation, fraud patterns, and volume forecasting on synthetic mobile payment transaction data, 6.36M transactions

Findings

1. Fraud is concentrated in two transaction types. TRANSFER carries a 0.77% fraud rate and CASH_OUT 0.18%; CASH_IN, PAYMENT, and DEBIT show zero labeled fraud in this data. Fraud TRANSFERs almost always drain the account: 96% of them take the origin balance to about zero in a single transaction, a sharper signature than the type-level rate alone suggests.
2. Account value is concentrated in half the base, split two ways rather than dominated by one tier. Quantile RFM segmentation splits accounts into 5 tiers; the two high-value tiers, frequent high spenders and high-value one-off senders, are 50% of accounts by count and hold 94.3% of total transaction value between them, close to evenly. The other 50% of accounts (low-value-frequent, standard, and dormant) account for just 5.7% of value.

Interventions

1. Real-time flag on near-total balance drain. Rather than a type-level rate threshold, flag TRANSFER transactions that take the origin account's balance to within 1% of zero in one step. That's the verified pattern behind 96% of labeled fraud TRANSFERs, and it's rare in legitimate traffic.
2. Tiered monitoring by RFM segment. Light-touch checks on high-frequency regulars, whose repeat behavior gives a stable baseline to compare against; tighter scrutiny on high-value transactions from accounts with little prior activity, which is the profile every labeled fraud CASH_OUT in this dataset matches (each is a single transaction from an account never seen before or since).

Method & data notes

- Segmentation uses quantile RFM (4 tiers on recency, frequency, monetary), not k-means. Faster, interpretable, and doesn't need a training or scoring pipeline.
- The forecast is a 7-day linear trend fit on the most recent week, not ARIMA or Prophet. Full-history volume shows a level shift roughly two-thirds through the window (early days run 5-14x later days), so fitting on the recent regime instead of the full history avoids extrapolating the early high-volume trend into a negative forecast.
- Checked for a literal TRANSFER-to-CASH_OUT mule account chain, a shared account ID linking a fraud transfer to a later cash-out, and found it's structurally absent: 0 accounts do both, and only 3 of 8,213 fraud rows share any account link at all. The two fraud types present as independent single-hop events in this file, not a traceable chain. Reported here since it contradicts a common assumption about this dataset.
- Merchant accounts (nameOrig/nameDest starting "M") carry a zero-balance placeholder for oldbalanceDest/newbalanceDest, a known data artifact excluded from balance logic rather than treated as a signal.
- Balance reconciliation (oldbalanceOrg - amount vs newbalanceOrig) mismatches on 79.8% of all rows. That's a data-generation characteristic of this dataset, not a fraud indicator, noted here so it isn't mistaken for one downstream.